

The + prescription for the p^+ integral

Is it legitimate here, and on what grounds

$$\kappa \equiv k \cdot (y' - z), \quad \xi \equiv p^+/k^+, \quad \lambda \equiv \Lambda/k^+, \quad \Xi \equiv (\vee - k^+)/k^+, \quad \ell_k \equiv \log \frac{k^+}{\Lambda}$$

0. Short answer

Yes — the + prescription is the right tool, and it is unusually clean here. But the reason you give for it is not the reason it works, and taken at face value that reason is false.

It works because, once the w' delta-function has been used, *the only p^+ dependence anywhere in the integrand is the phase $e^{-i\frac{p^+}{k^+}k \cdot (y' - z)}$* . Wilson lines, colour sources and all four kernels are p^+ -independent, so the subtraction term is simply “the same integrand with the phase set to 1”.

It does **not** work because “there is no transverse divergence”. That statement was established *with the phase present*. At $p^+ = 0$, where you do the subtraction, the phase is gone and the subtraction term **is** transversely logarithmically divergent — see §3. That is precisely why this row on its own produces ℓ_k^2 and not ℓ_k .

1. First, a correction

In the previous note I flagged the overall $\frac{1}{k^+}$ as possibly counted twice. **That flag was wrong.** Your Eq. (3) settles it, and your w' step is correct.

The delta sets $w' = y' + \frac{p^+}{k^+}(y' - z)$, so $w' - z = \frac{p^+ + k^+}{k^+}(y' - z)$ and the first kernel supplies $\frac{k^+}{p^+ + k^+}$. Against the $\frac{1}{k^{+2}}$ in Eq. (2), and pulling $(p^+ + k^+)$ out of the bracket,

$$\frac{1}{k^{+2}} \cdot \frac{k^+}{p^+ + k^+} \left[\delta_{k'm} \delta_{ij} - \frac{p^+ + k^+}{p^+} \delta_{jm} \delta_{ik'} - \frac{p^+ + k^+}{k^+} \delta_{im} \delta_{jk'} \right] = \frac{1}{k^+} \left[\frac{\delta_{k'm} \delta_{ij}}{p^+ + k^+} - \frac{\delta_{jm} \delta_{ik'}}{p^+} - \frac{\delta_{im} \delta_{jk'}}{k^+} \right]$$

identically (difference checked symbolically to be 0). And the measure does not eat that $\frac{1}{k^+}$: every term of the bracket carries exactly one inverse power of momentum, so with $dp^+ = k^+ d\xi$ the k^+ cancels *inside* the bracket,

$$\int dp^+ \frac{1}{p^+ + k^+} = \int \frac{d\xi}{1 + \xi}, \quad \int dp^+ \frac{1}{p^+} = \int \frac{d\xi}{\xi}, \quad \int dp^+ \frac{1}{k^+} = \int d\xi.$$

So $\int dp^+ [\text{bracket}] = \int d\xi \left[\frac{A}{1+\xi} - \frac{B}{\xi} - C \right]$ with the $\frac{1}{k^+}$ surviving in front, and the p^+ -integrated expression you wrote earlier is correct as it stands.

2. The prescription, and what it needs

Only the middle term has a pole; $\frac{1}{p^+ + k^+}$ is regular at $p^+ = 0$ and $\frac{1}{k^+}$ has no pole at all. For that one term the identity is

$$\int_{\Lambda}^P \frac{dp^+}{p^+} g(p^+) = \underbrace{\int_0^P \frac{dp^+}{p^+} [g(p^+) - g(0)]}_{\text{finite}} + g(0) \log \frac{P}{\Lambda} + O(\Lambda), \quad P = \vee - k^+,$$

equivalently, as a distribution in ξ ,

$$\frac{1}{\xi} \longrightarrow \left[\frac{1}{\xi} \right]_+ + \delta(\xi) \log \frac{\Xi}{\lambda}, \quad \int_0^{\Xi} d\xi \left[\frac{1}{\xi} \right]_+ h(\xi) \equiv \int_0^{\Xi} d\xi \frac{h(\xi) - h(0)}{\xi}.$$

Two things it requires, and they are different:

1. $g(0)$ **must exist**. Here it does, trivially: the phase is the only p^+ dependence, so $g(0)$ is the integrand at $\xi = 0$, phase replaced by 1. This is the part of your reasoning that is right, and it is what makes the subtraction *local* in the transverse variables.
2. $\int d^2(\text{transverse}) g(0)$ **must converge**, or you have merely moved a divergence around. This is the part that fails — see §3.

It reproduces the Ei answer exactly

With $g = e^{-i\kappa\xi}$ the subtracted integral is $\text{Ei}(-i\kappa\xi) - \gamma_E - \log(i\kappa\xi)$, and adding $\log \frac{\Xi}{\lambda}$ gives back $\text{Ei}(-i\kappa\xi) - \text{Ei}(-i\kappa\lambda)$ because $\text{Ei}(-i\kappa\lambda) = \gamma_E + \log(i\kappa\lambda)$:

```
kap=+1.7 lam=1e-9: Ei form (19.6162521668 - 1.55492578417j) + form (19.6162521668 - 1.55492578587j)
kap=-0.4 lam=1e-9: Ei form (21.1178768959 + 1.60850528357j) + form (21.1178768959 + 1.60850528397j)
kap=+5.0 lam=1e-9: Ei form (18.5385131804 - 1.57585619805j) + form (18.5385131804 - 1.57585620305j)
```

Same numbers, different bookkeeping. So the $+$ prescription is not an approximation — it is an exact rearrangement of the integral you already have.

3. The catch: $g(0)$ is transversely divergent

The pole term is $-\frac{\delta_{im}\delta_{jk'}}{p^+}$, whose contraction on the four kernels leaves, for the z integration,

$$\frac{(y' - z) \cdot (x - z)}{(y' - z)^2 (x - z)^2} = \mathcal{K}(x, y'; z),$$

the **real-emission piece of the BK kernel**. It falls only as $1/z^2$, so $\int \rho d\rho \rho^{-2}$ is logarithmically divergent. For $p^+ \neq 0$ the phase $e^{-i\frac{p^+}{k^+}k \cdot (y' - z)}$ cuts that log at $|z| \sim k^+/(p^+|k|)$ — which is exactly why the earlier analysis found no transverse divergence. At $p^+ = 0$ the cut is removed.

Measured angular averages, $\rho^2 \langle \cdot \rangle_\theta$ at large ρ :

ρ	real piece $-2\mathcal{K}$	full BK $\mathcal{M}$
10^1	-1.993815	0.016466
10^2	-1.999938	0.000164
10^3	-1.999999	0.000002
10^4	-2.000000	0.000000

So the two divergences are **entangled**: the rapidity pole $\frac{1}{\xi}$ multiplies a transverse logarithm $\log \frac{1}{\xi}$, and that is where ℓ_k^2 comes from. A $+$ prescription applied now does not remove it — it just parks (divergent transverse integral) $\times \log \frac{\sqrt{-k^+}}{\Lambda}$ in the subtraction term.

4. What makes it legitimate

Add the virtual partners of the same order, so that the real piece is completed to the full BK kernel,

$$\mathcal{M}(x, y'; z) = \frac{(x - y')^2}{(x - z)^2 (y' - z)^2} = \frac{1}{(x - z)^2} + \frac{1}{(y' - z)^2} - 2\mathcal{K}(x, y'; z),$$

whose large- $|z|$ angular average vanishes at order $1/\rho^2$ ($1 + 1 - 2 = 0$ in the table above) and which therefore falls as $1/\rho^4$. **Then $g(0)$ is transversely finite**, and the $+$ prescription does exactly what you want:

$$\int_{\Lambda}^P \frac{dp^+}{p^+} g = \underbrace{\log \frac{\sqrt{-k^+}}{\Lambda} \times \mathcal{M} \otimes (\text{LO})}_{\text{one rapidity log; absorb into JIMWLK/BK}} + \underbrace{\int_0^P \frac{dp^+}{p^+} [g(p^+) - g(0)]}_{\text{NLO impact factor, finite}}$$

and the answer carries a *single* logarithm, as an NLO rapidity subtraction must.

A scheme point worth settling now

The prescription isolates $\log \frac{\sqrt{-k^+}}{\Lambda}$, **not** $\ell_k = \log \frac{k^+}{\Lambda}$. The difference $\log \frac{\sqrt{-k^+}}{k^+}$ is finite and purely a choice of where the evolution stops and the impact factor begins — but it is a large number in practice, and it must match the convention used for the initial condition of the evolution equation:

```
V = 100 k+ :   Lambda/k+ = 1e-9:   log(Xi/lam) = 25.31839   l_k = 20.72327   difference = 4.59512
               Lambda/k+ = 1e-6:   log(Xi/lam) = 18.41063   l_k = 13.81551   difference = 4.59512
```

Subtracting at $\xi = 0$ with the upper limit Ξ is the natural choice for the $+$ distribution; subtracting with ℓ_k is the natural choice for matching to BK. Pick one and carry the finite difference explicitly.

5. Recommended order of operations

1. Collect the virtual diagrams so the z -integrand of the pole term is $\mathcal{M}$, not $-2\mathcal{K}$. Until this is done, no rapidity subtraction is well defined.
2. Then apply the $+$ prescription to $\frac{1}{p^+}$ only; leave $\frac{1}{p^++k^+}$ and $\frac{1}{k^+}$ alone (no poles).
3. Check that the coefficient of the isolated log is $\mathcal{H}_{\text{JIMWLK}} \otimes \frac{d^3 N_{\text{LO}}}{d^3 k}$ after the colour structure is reduced to dipole/quadrupole operators. If it is not, something upstream is still missing.
4. Absorb the log into the target weight functional at rapidity Y , fixing the finite offset of §4 by the convention of the evolution equation you are matching to.
5. The remaining $\int_0^P \frac{dp^+}{p^+} [g(p^+) - g(0)]$ is the NLO impact factor; do its transverse integrals with the master $B^{mj}(Q, c)$.

`zint/src/plusprescription.py` — every number above, written to `zint/RESULTS_plus.txt`.